

**THE UNIVERSITY OF HONG KONG**  
**FACULTY OF BUSINESS AND ECONOMICS**  
**[ECONXXXX] AI-Enabled Economic Policy Research**

**GENERAL INFORMATION**

**Instructor:** Prof. Michael B. Wong

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**Consultation times:** TBD

**Tutor:** TBD

**Pre-requisites:** An introductory course in econometrics or statistics. Comfort with at least one programming language (R, Python, or Stata).

**Co-requisites:** None

**Mutually exclusive:** None

**Course Website:** HKU Moodle

**Reference Books:** Bardach & Patashnik (2019), A Practical Guide for Policy Analysis; Bellemare (2022), Doing Economics

**The following programming languages are used in this course:** R, Python, or Stata (student choice)

VS Code with GitHub Copilot Pro+ is also recommended.

**COURSE DESCRIPTION**

This is a five-week intensive course built around a single team research project in economic policy. Students form teams of three, identify a policy question relevant to Hong Kong or the broader region, assemble data, conduct empirical analysis, and produce a 10-page policy research paper. Structured lectures introduce the fundamentals of policy-oriented research: how policy questions differ from academic research questions, how to bring data to bear on policy decisions, and how to write for decision-makers rather than referees. Students are strongly encouraged to use AI tools (LLMs, AI coding assistants) at every stage. The course is project-driven: four graded milestones (proposal, peer review, presentation, final paper) compress the full research cycle into five weeks.

**COURSE OBJECTIVES**

This course aims to develop students' ability to conduct data-driven policy research and communicate findings effectively. Students will learn to formulate policy questions, design and execute appropriate empirical analyses (from descriptive statistics to data visualization), write for policy audiences, and give and receive constructive feedback. The course also equips students to use AI tools productively in their research workflow.

**Programme Learning Outcomes**

**PLO1: Understanding of fundamental theories and new development in economics:** Students should acquire the fundamental concepts and the cutting-edge insights in economics. They will be able to use these theoretical knowledge and tools to analyze economics problems and develop solutions for them.

**PLO2: Mastering of skills in analyzing economic data:** Students should be able to use the appropriate tools and frameworks to analyze economic data, and be able to make use of the analysis results to develop valuable insights for different economic issues.

**PLO3: Demonstration of ability to apply economic knowledge and analytical skills to address policy and business problems with a global outlook:** Students should be able to apply and integrate economic knowledge and analytical skills to identify and tackle policy and business problems with global vision.

**PLO4: Awareness of ethical concerns in economic issues:** Students should demonstrate the capacity to recognize, understand and develop appropriate options in ethical and social issues in economics. They are to adhere to and respect university regulations on academic conduct.

**PLO5: Mastering of communication skills:** Students should be able to articulate in public and be able to make convincing and coherent presentations, write effectively and use specific technical terminology in contemporary presentation tools context.

### COURSE LEARNING OUTCOMES

| Course Learning Outcomes                                                                                                                                              | Aligned PLOs |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| CLO1: Identify and frame a policy question that is specific, empirically tractable, and decision-relevant.                                                            | PLO1, PLO3   |
| CLO2: Design and execute a data analysis appropriate to a policy question — whether descriptive, comparative, or regression-based — and present the results honestly. | PLO2, PLO3   |
| CLO3: Write a policy research paper that communicates findings clearly to a non-specialist audience while maintaining analytical rigor.                               | PLO3, PLO5   |
| CLO4: Give and receive constructive feedback on policy research in progress, and use AI tools productively in the research workflow.                                  | PLO4, PLO5   |

### COURSE TEACHING AND LEARNING ACTIVITIES

| Course Teaching and Learning Activities                                   | Expected Hours | Study Load (% of study) |
|---------------------------------------------------------------------------|----------------|-------------------------|
| T&L1. Interactive lectures (Sessions 1–4)                                 | 12             | 9%                      |
| T&L2. Peer review workshop (Session 5)                                    | 3              | 2%                      |
| T&L3. Team working sessions / lab (Sessions 6–7, 10)                      | 9              | 7%                      |
| T&L4. Team presentations (Sessions 8–9)                                   | 6              | 5%                      |
| T&L5. Team project work (outside class): data analysis, writing, revision | 70             | 54%                     |
| T&L6. Reading and self-study                                              | 30             | 23%                     |
| <b>Total</b>                                                              | <b>130</b>     | <b>100%</b>             |

### ASSESSMENT METHODS

| Assessment Methods | Brief Description | Weight | Aligned CLOs |
|--------------------|-------------------|--------|--------------|
|--------------------|-------------------|--------|--------------|

|                                 |                                                                        |             |                  |
|---------------------------------|------------------------------------------------------------------------|-------------|------------------|
| A1. Participation               | Engagement in lectures, workshops, presentations, and lab sessions     | 10%         | CLO1, CLO4       |
| A2. Proposal (2 pages)          | Team policy research proposal: question, context, data, approach       | 15%         | CLO1, CLO2       |
| A3. Peer review report (1 page) | Structured critique of another team's proposal                         | 10%         | CLO4             |
| A4. Presentation (10–15 min)    | Team policy briefing with Q&A                                          | 15%         | CLO1, CLO2, CLO3 |
| A5. Final paper (10 pages)      | Policy research paper: question, data, analysis, results, implications | 50%         | CLO1, CLO2, CLO3 |
|                                 | <b>Total</b>                                                           | <b>100%</b> |                  |

## STANDARDS FOR ASSESSMENT

### Course Grade Descriptors

| Grade     | Descriptors                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A+, A, A- | <ul style="list-style-type: none"> <li>• Policy question is specific, well-motivated, and clearly decision-relevant</li> <li>• Data analysis is appropriate, clearly explained, and honestly presented</li> <li>• Paper is well-written, accessible to non-specialists, with strong policy implications</li> <li>• Presentation is polished; handles Q&amp;A with confidence and insight</li> <li>• High engagement in peer review and class discussions</li> </ul>       |
| B+, B, B- | <ul style="list-style-type: none"> <li>• Policy question is identifiable and relevant, though framing could be sharper</li> <li>• Data analysis is reasonable with minor gaps; results presented clearly</li> <li>• Paper is well-organized and readable; policy implications are present but could be more specific</li> <li>• Presentation is competent; Q&amp;A responses are adequate</li> <li>• Good engagement in peer review and class discussions</li> </ul>      |
| C+, C, C- | <ul style="list-style-type: none"> <li>• Policy question exists but is vague or only loosely connected to the analysis</li> <li>• Data analysis is basic; some errors or omissions in presentation of results</li> <li>• Paper is understandable but disorganized or overly technical for the intended audience</li> <li>• Presentation covers the material but lacks clarity or confidence</li> <li>• Some participation in peer review and class discussions</li> </ul> |
| D+, D     | <ul style="list-style-type: none"> <li>• Policy question is unclear or has no meaningful policy relevance</li> <li>• Data analysis has significant errors or is inappropriate for the question</li> <li>• Paper is difficult to follow with major structural problems</li> <li>• Presentation is poorly prepared; unable to answer basic questions</li> <li>• Minimal participation in class activities</li> </ul>                                                        |
| F         | <ul style="list-style-type: none"> <li>• No identifiable policy question or contribution</li> <li>• Data analysis is absent, wrong, or misleading</li> <li>• Paper fails to meet minimum standards of academic writing</li> <li>• Presentation is absent or entirely unprepared</li> <li>• No meaningful participation; failure to attend required sessions</li> </ul>                                                                                                    |

| COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE |      |               |                                                          |                 |
|------------------------------------------------|------|---------------|----------------------------------------------------------|-----------------|
| Session                                        | Week | Format        | Topic                                                    | Milestone       |
| 1                                              | 1    | Lecture       | What is a policy question?                               | —               |
| 2                                              | 1    | Lec + Lab     | AI tools orientation; team formation & topic exploration | —               |
| 3                                              | 2    | Lecture       | From policy question to data analysis                    | —               |
| 4                                              | 2    | Lec + Lab     | Writing for policy audiences; proposal workshop          | Proposal due    |
| 5                                              | 3    | Workshop      | Peer review workshop                                     | —               |
| 6                                              | 3    | Lab           | Team working session: analysis & writing                 | Peer review due |
| 7                                              | 4    | Lab           | Team working session: results & draft preparation        | —               |
| 8                                              | 4    | Presentations | Team presentations (Groups 1–6)                          | Presentation    |
| 9                                              | 5    | Presentations | Team presentations (Groups 7–12)                         | Presentation    |
| 10                                             | 5    | Lab           | Final working session: revision & submission             | Final paper due |

| REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Bardach, E. and Patashnik, E. (2019). A Practical Guide for Policy Analysis. 6th ed. CQ Press.</p> <p>Bellemare, M. (2022). Doing Economics: What You Should Have Learned in Grad School — But Didn't. MIT Press.</p> <p>Shapiro, J. (2022). "How to Give an Applied Micro Talk." Unpublished guide.</p> <p>Gentzkow, M. and Shapiro, J. (2014). "Code and Data for the Social Sciences: A Practitioner's Guide."</p> <p>Korinek, A. (2023). "Generative AI for Economic Research." Journal of Economic Literature.</p> <p>Additional readings and materials will be posted on Moodle.</p> |

| MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE                                                              |
|-------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Online response via Moodle site</li> <li>• Others: SFTL</li> </ul> |

| COURSE POLICY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>AI Use Policy:</b> Students are strongly encouraged to use AI tools (LLMs, AI coding assistants, AI-powered search tools) at every stage of their project. There is no restriction on which tools are used or how they are used. The only requirement is that students must understand and be able to defend everything in their paper. Undocumented reliance on AI-generated content that a student cannot explain constitutes academic dishonesty.</p> <p><b>Teams:</b> Teams of 3 are required. Teams of 2 or 4 require explicit approval from the instructor and are typically granted only when there is a shortage or excess of students.</p> |

**Academic Honesty:** An orderly learning environment is extremely important. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.

**Attendance:** Attendance at all 10 sessions is expected. This is a compressed, project-driven course; missing sessions directly impacts team performance.

#### ADDITIONAL COURSE INFORMATION

All milestones are submitted via Moodle. Late submissions will incur a penalty of 10% per day.

Lecture materials and assignment instructions will be posted on Moodle.

Recommended tool setup: VS Code with GitHub Copilot Pro+ (\$39/month). Setup instructions will be provided in Session 2.